

CHAPTER ELEVEN

Introduction to the Base Period

As we mentioned earlier, the sports of mountain running and skimo are complementary both from the seasonal standpoint and because most of the physiological demands are the same. Their training methods share a great deal in common. So it makes sense to address their training in one book. The difference in the approach you choose will depend only on whether you want to focus just on running, just on skimo, or both. We'll cover all three options in the remaining chapters.

Well over 80 percent of your time and energy expenditure are consumed on the uphill sections for skimo and ski mountaineering, and more than 50 percent for running. Even a few percent improvement in the uphill speed can easily translate into several minutes gained over the course of a long climb. Hence, much of the focus here will be on improving uphill performance. The aerobic base miles (or kilometers) that a runner puts in have their analog in skimo as base vertical feet (or meters). For skiers this means accumulating many thousands of feet (or meters) of Zones 1–2 aerobic work supplemented by some strength and higher-intensity training. While runners' training also needs to include sufficient elevation gain and loss, it also has to prepare them to cover the required distance of their goal event. The takeaway is that the bulk of the training is essentially the same for both sports: time on your feet going up and down mountains has to form the vast majority of your training if you hope to be well prepared for the race season. The following information is intended to help you develop a training plan that suits your goals while considering your training history.

The purpose of the Base Period is to increase your work capacity and improve your fatigue resistance. You build this fitness base so that you can handle more

work and eventually more intense work, which prepares you for the races themselves. You'll do this by using the progression principle that we discussed in earlier chapters: slightly exceeding your work capacity in many workouts, then recovering and repeating this process for months at a time. The progression must be gradual (so your body can adapt) and continuous (without breaks). It underlies the entire plan and its effectiveness. Exercising in a random and sporadic way will not give you the best results.

During much of the Base Period, you will be a little too tired to perform at your best. This is why elite athletes build their base *before* the main competitive period. If you do insert races or significant training events into your Base Period, they should be low priority (C level) and barely exceed what you are typically doing in training. These will then minimally disturb the overall training progression toward your ultimate goal. Recall from the capacity versus utilization discussion (see page 74) that it is just not possible to effectively build capacity (fitness) while trying to maximally utilize it (as you will during races or important events) at the same time. Waiting until December to begin a skimo training program that has to overlap with the racing season will certainly compromise either the training or the race results, or both.

All the plans discussed in this book assume that you want to maximize your performance and are willing to undertake a focused plan for months on end. Neither this book nor the plans in it are intended for those looking for shortcuts.

There are no shortcuts.

We want to once again emphasize the principles that you need to follow: maintaining *consistency* in your training routine while *gradually* increasing the training load and *modulating* the training from hard to easy over days and weeks.

The Low-Hanging Fruit

Train your weaknesses and race your strengths.

– Eddie Borysewicz (famed cycling coach)

As with all training, the Base Period must be individualized to best meet your needs. Heeding Eddie B's sage advice, you need to figure out: What are your weaknesses? What is holding you back from improving? Doing this requires detached analysis coupled with some knowledge. Imparting this knowledge to you so that you can make better-informed decisions is why we focused in the early sections of this book on the fundamentals of training for endurance.

We're going to give you a few tips on how to pick out your weaknesses in a minute, but one sure way *not* to do this is to do in training only the things you are good at or enjoy. What's wrong with that approach? Plenty. We all gravitate



Alyson Dimmitt Gnam logging miles on alpine singletrack surrounded by blooming bear grass in Glacier National Park, Montana. Photo: Steven Gnam

toward doing things we are good at in most areas of our lives. We also avoid things we don't enjoy or are worse at—nobody likes to do things they suck at! This natural tendency often drives us to overemphasize in training the areas where we are already the strongest while avoiding the things where we have more room to improve. Continuing to make significant gains in your strong areas, year after year, is going to be tough, whereas a shift of training emphasis to your areas of weakness can deliver big results. Here are a couple of common examples:

- If you are stronger (relative to your competitive standing) on the uphill in running races but get dropped on flatter and faster terrain, this would indicate that your aerobic endurance is probably relatively higher than your running economy. A wise training trade-off for this type of runner will be to shift some emphasis to flatter, faster running and technique work.
- If you have a strong downhill skiing background and manage to dominate on the downhill sections of skimo races but lose time on the uphills, then you'd be wise to put more effort into increasing your aerobic endurance.
- If your legs get very sore and your muscle fatigue lingers for several days after a long run or race with a lot of vertical gain and loss, you may need to increase your leg strength and muscular endurance in the Base Period.



Luke Nelson's feet after running The Rut 50K race in Big Sky, Montana. Photo: Steven Gnam

There is no better proving ground in sports than competition. Examine your racing history: On what parts of the course, or even on what type of courses, are you relatively better or worse than your competitor peer group? The areas where you are worse than your peer group should be the areas you pay more attention to. Beginners in a sport usually make rapid improvements in all areas and quickly see big performance gains. But, the better you get, the more elusive those gains will become.

Even professional athletes have limited energy and time. Most amateurs will have quite restricted time and energy constraints for training. They especially need to make wise use of their training time. The message here is that just running more miles or putting in more vertical meters, while certainly important, may at some point begin to show diminishing returns on the training investment, and the savvy coach and athlete need to be open to changing the training stimuli in order to continue making gains.

The Cornerstone of the Base

For most readers and for all the sports covered in this book, the number one area of training focus must be on increasing that all-important aerobic base. Without

this base all the rest is mere window dressing. For seasoned endurance athletes with that big base of aerobic work, the training can and must take on multiple dimensions and will therefore become more complex. If you are nodding in recognition as you read the first “low-hanging fruit” example above, you need to recognize that your aerobic endurance is not what is holding you back in races and be brave enough to step away from the safety of only training your strength. This doesn’t mean that you stop the aerobic training, but it does mean a re-allotment of training time to include the faster flat running.

While there is no simple rule of thumb for deciding how much time to spend working on running economy or skimo technique, deciding if your aerobic base is big enough for the current Base Period and when it is most beneficial to include higher-intensity training is a much simpler task.

The best way to decide how much emphasis you still need to place on basic aerobic fitness versus adding more high-intensity endurance training is to refer back to the Ten Percent Test (see page 91). If the difference between your AeT and LT (in terms of heart rate) is 10 percent or less, you should include up to two weekly high-intensity aerobic endurance sessions in your Base Period. If your AeT-to-LT spread is greater than 10 percent, delay the introduction of Zone 3 workouts and limit the higher-intensity workouts to no more than once a week.

The reason for this distinction is that those with a higher basic aerobic work capacity can handle more intensity, meaning they will recover and benefit better from it. Those with a wider spread still have some low-hanging aerobic fruit to harvest and, while they can surely benefit from the addition of high-intensity training, their gains will never be as great nor as long lived as for those who have more aerobic capacity.

The strength training in this period plays only a supporting (albeit important) role for the other more sport-specific training. If you do find yourself lacking the basic strength as determined in the strength assessment test in the chapter 7, then you should start with remedial strength training. Those with a better strength base will find the Specific Strength recommendations useful. But you won’t be spending many hours each week in a gym lifting weights in any of our programs.

The Mechanics of Planning

We’ve gone to great lengths to explain the need for individuality and how you can adjust and monitor your training to give the best results. We emphasized the nuanced gray areas that make coaching more art than science. Now we are about to embark on what we referred to earlier as the “mechanistic” approach. We’re going to describe ways to lay out a long-term plan for yourself. Knowing full well that, no matter how pure your intentions, you *will* deviate from this, or any, long-term plan.



ATHLETE STORIES

Lining Up and Knowing I Would Win

Rico Elmer

I was born in a small valley near the Linth River. Though the mountains there are not high—around 10,000 feet—the valley floor is at 1,000 feet and the summits form a cul-de-sac that traps the moist air coming from the west, so we always have a lot of snow. I started cross-country skiing early, as most kids in the area did. I participated in training and racing from an early age. I raced at the World Cup level in cross-country, but I never really felt at home there. I was not the most stellar technician, and I didn't do well in any important races except when conditions were horrible.

I did my first ski mountaineering races in my early twenties, and when I was twenty-five I started training with a coach, Gregor Hagmann.

At that time the Swiss had a system where an athlete could apply for employment with the Swiss Border Guard and be paid a salary, yet only need to train ... and show results. When I was twenty-six I got this job and I started training full-time. The results came: when I was twenty-nine I won my first major race, the Patrouille des Glaciers. My teammates, Emanuel Buchs and Damien Farquet, and I set a course record.

In those days in the sport of ski mountaineering, there were not too many transitions, usually only one to three per race, and there was a lot of flat skinning. My experience cross-country skiing was valuable in that regard, because I knew how to glide a ski. My cross-country ski training in my youth helped me a lot.

In my first years of training with Gregor I could really feel that I was improving significantly from year to year. Then from 2000 to 2004, I knew at each starting line that, as long as I didn't make a big mistake, I was going to win the race. During that time I didn't feel

I was really improving, I just wanted to hold onto that level as long as I could. In 2006, I felt that I started to lose that really good fitness, that I had been in the sport too long. I was growing tired of it.

My summer training was always a lot of volume. On a typical day I'd wake up and go for a twenty- to thirty-minute run, come back, have breakfast, and then go for my long run of the day. After lunch I'd take a nap, and then go for a shorter one- to two-hour run in the afternoon before doing some Core Strength training. I often mixed running with roller skiing. Roller skiing is excellent training for ski mountaineering. I did a lot of double-poling on the roller skis.

It's my belief that having a big base is key to succeeding on race day, though there are other puzzles to unlock when it comes to training—including intervals. Intervals were always an important part of my progression. At the end of a race season I would take one to two months completely off from training, and then as the snow started to melt, I'd start to run again. After two to three months of volume, mostly mountain running and roller skiing, I'd start to do intervals. I did all kinds of intervals: 30/30s, pyramids, frog-hops front and back, everything Gregor could invent. I often did intervals during my long workout; never just on a whim though, it was all planned.

In the fall I always did some kind of race—a mountain bike race or mountain running race, but not a ski race—to help get my mind into racing mode, into going as fast as possible.

Eating was also very important. Not to be crazy about nutrition, but to eat enough. I experimented with getting lighter. I am five-foot-ten, and my best

weight was 154 pounds. I went down to 143 pounds at one point, but I wasn't nearly as fast and I was getting sick all the time. I ate a balanced diet while training; I always eat some of everything.

I followed the same routine before each race. If the race was on Saturday, on Wednesday I would do one interval workout, hard but short. Thursday I would do about twenty easy minutes on a bike and then travel to the race. Friday I would go out on my skis and check out the course. During this check I would do two to three quick sprints, each a bit harder, with the last one to get my heart to max, just for one minute or less. The night before the race I ate spaghetti with olive oil, just plain. Three hours before the race I ate muesli or bread, and then fifteen minutes before the race I ate just a little carbohydrate again. I could eat almost anything, as long as it was healthy, the morning before a race.

To have a good race, I had to be nervous. At some races I was so nervous my fingers tingled. That was too much. But I always prioritized my races, focusing on one or two each year. ■

Rico Elmer is a Swiss ski mountaineer who dominated the ski mountaineering race circuit in the early 2000s: he won the European Championship of Ski Mountaineering in the team division in 2003 (with Damien Farquet), and he won the individual title at the World Championship in 2004. He has won nearly every major race in the ski mountaineering world including the Patrouille des Glaciers (1998 and 2000) and the Trofeo Mezzalama (2003). Rico is the former head coach of the Swiss National Ski Mountaineering Racing Team; he coached six-time world (team and individual) champion Florent Troillet.



Tadej Pivk during the Dolomites Sky Race on Sass Pordoi, Italy. Photo: Federico Modica

To reduce the chances for significant deviations and drastic changes to this plan, we've developed some ways that allow you, by referring back to the previous chapters, to build a fairly customized plan and then to make the inevitable adjustments on the fly.

As the earlier sections of this book explained, there is no one-size-fits-all approach to training. Trying to blindly copy the training of a highly successful athlete is almost guaranteed to lead to disappointment and even worse.

To guide you through the mechanics of creating a plan, refer first to the flow-chart below.



Fig. 11.1 The Mechanics of Planning

First select the athlete category that most closely fits your circumstances. The macro plan you assemble in step 2 will help you have an overview of the coming weeks and months without getting mired down by planning the actual days. Use step 3 to create a general daily structure to the weeks. Lay out actual workouts in the weekly structure only one to two weeks ahead. This will save you frequently changing the plan based on unforeseen circumstances and help that "blind adherence" to the plan that we cautioned against earlier.



Federica Biofava charges up a hillside in the Camonica Valley of northern Italy. Photo: Davide Ferrari

Step 1. Choose Your Athlete Category

To start this process, you need to honestly assess what you are bringing to this project.

The needs of the experienced mountain athlete with several seasons of training and multiple races in the bag are quite different from what is needed by those just coming into these sports with the hopes of completing their first 50K running race or trying out a skimo race or two next season, or someone who may have never followed any sort of training program but wants to make the most of a once-in-a-lifetime ski tour. Select from one of the two categories that fits you best.

- The first, Category 1, is for the beginner and less experienced athlete who needs to *build* a base of fitness.
- The second, Category 2, is for athletes with at least one successful season (preferably several seasons) of training behind them who need to *maintain* and supplement an already-existing base.

CATEGORY 1

Athletes in need of building a base:

- Have not averaged more than 400 hours per year of focused training for running or skimo during the past two years.

or

- Are under the age of eighteen.

or

- Have not, during the previous twelve months, successfully* completed races or events at the distance/duration they are aiming for in the next training period.

If you fit into this category, the single most effective thing you can do is to increase the volume of training you are doing. Ideally you will do this over a long period in your off-season.

Note: If you are training at well below 400 hours per year, then you should expect to spend additional training cycles in this base-building phase of training. This applies especially to juniors and the inexperienced. Do not skip to the Category 2 section too early or you will struggle to handle the workload and may not see improvement.

*Successfully: This means you performed well and you were happy with the result. You achieved or came close enough to the goals you had set for yourself that you were satisfied. You were not destroyed or injured by the event.

CATEGORY 2

Athletes in need of maintaining their base while supplementing it:

- Have trained consistently at over 400 hours per year for at least the past two years. This athlete never really stops training other than short breaks, so his or her base remains high.

and

- Are over the age of eighteen.

and

- Have, during the previous twelve months, successfully* completed races or events at or near** the distance/duration they are aiming for this training period.

and

- Race several to many times during one year.

If you fit into this group, increasing the specificity of your training will yield the biggest gains. This means more training that mimics the demands of the event itself and, to the extent that it does not reduce your Zones 1–2 base training volume appreciably, more high intensity.

*Successfully: This means you performed well and you were happy with the result. You achieved or came close enough to the goals you had set for yourself that you were satisfied. You were not destroyed or injured by the event.

**Near: This means you have successfully completed a race or event that sits in the conventional progression of difficulty or distance as the next step below the targeted event for this training cycle.

Step 2. Select a Macrocycle Plan

In this section we'll lay out a few admittedly idealized (plans rarely work out as planned), longer-term plans such as the ones below. Select the type of plan from this group that best fits your goals and individual needs. We've tried to provide a variety of plans so that you can find one that is right (or close to right) for you. This system allows you the maximum freedom to structure your own plan while giving guidance to help you apply sound training practice to the organization of that plan. This particular chart shows the training progression using distance as a proxy for training load for an athlete targeting his or her first 50K ultrarun. The blue bars represent the weekly distance covered in miles.

Above each week is a letter: B (Base), I (Intensity), R (Recovery), S (Specificity), T (Taper), and G (Goal). These letters indicate the week type. You can start to build your plan in a spreadsheet, calendar, or notebook by transferring the workouts from the menu for the appropriate type of week into the days of week.

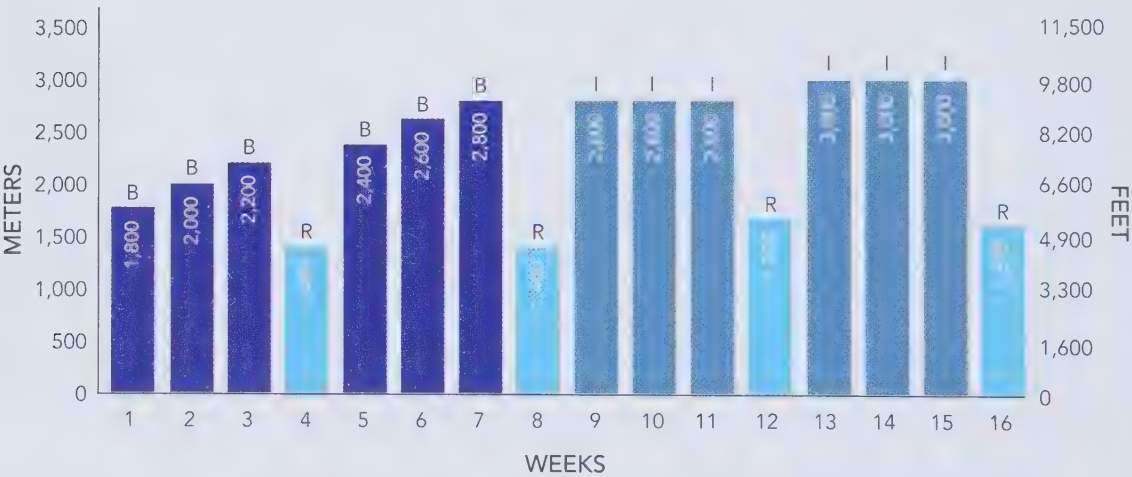


Fig. 11.2 Preseason Skimo Progression, Cat 1 First Season Athlete

This plan is for someone interested in trying his or her first couple of skimo races this season. You should have an average weekly vertical of around 1,500–1,700 meters for the previous eight weeks to support this plan. If your weekly average is lower, reduce week 1 vertical by your percent below 1,600, then continue the progression percentage as shown.

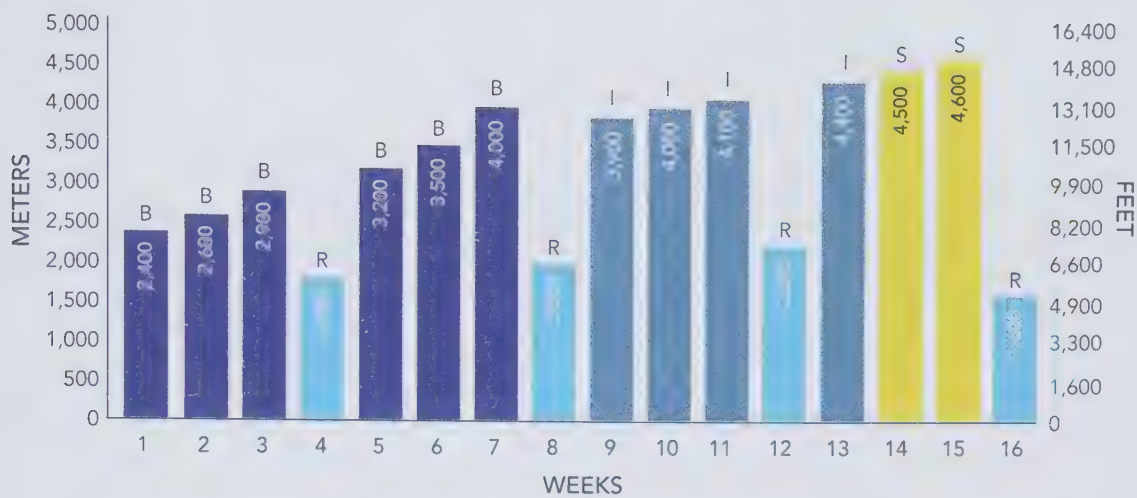


Fig. 11.3 Early-Season Skimo Progression, Cat 2 Athlete

This is for the athlete making the transition from a summer of mountain running who has averaged a weekly vertical of around 2,000-2,200 meters for the previous eight weeks. The plan is meant to end as the full multirace season begins. This plan could also work well for a ski mountaineering goal such as a hut trip or a grand tour like the Haute Route.

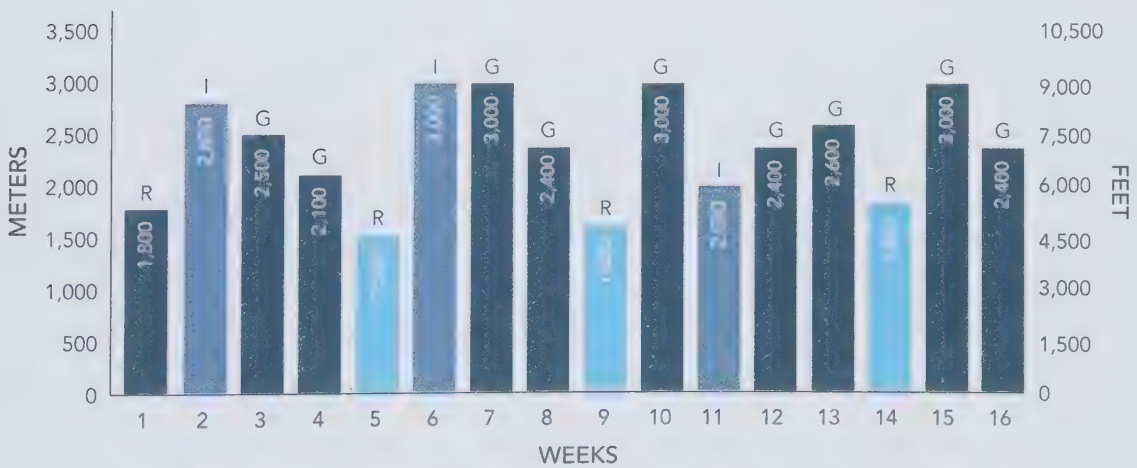


Fig. 11.4 Skimo Race Season Progression, Cat 2 Athlete

This plan is for someone competing in a race series extending from December until April. This plan should be preceded with a plan following close to Figure 11.5.



Fig. 11.5 Actual Skimo Training Progression, Cat 2 Athlete

This is an actual training plan for a nationally competitive skimo racer with an end-of-season A priority race in March doing a few C category races for training during the winter. The actual training will probably not follow the plan exactly. Regardless, this athlete achieved personal best results throughout the season. This progression starts in mid-October after a summer of base training. Could be used for races like Patrouille des Galciers and Power of Four.



Fig 11.6 Progression in Vertical, Hut-to-Hut Ski-Touring Trip

Here is an example of a progression in vertical meters for someone hoping to comfortably handle a multiday hut-to-hut ski touring trip such as the Haute Route or Wapta traverse. This would be the bare minimum for a trip such as the Hoch Tirol.

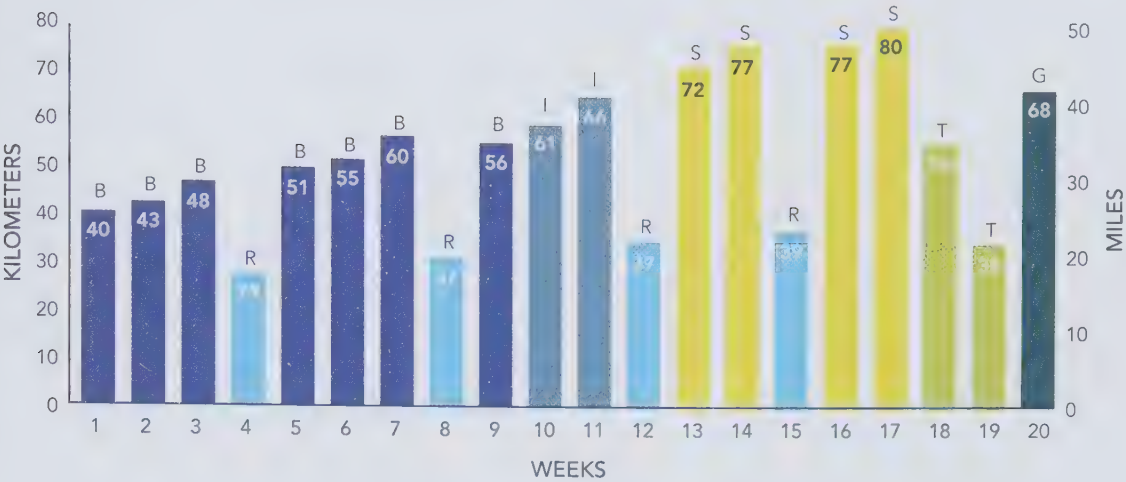


Fig. 11.7 Idealized Plan, Cat 1 First Time 50K Running Race
This is for an athlete targeting his or her first 50K running race. It could also serve as a program for a recreational runner with a personal goal of similar distance.

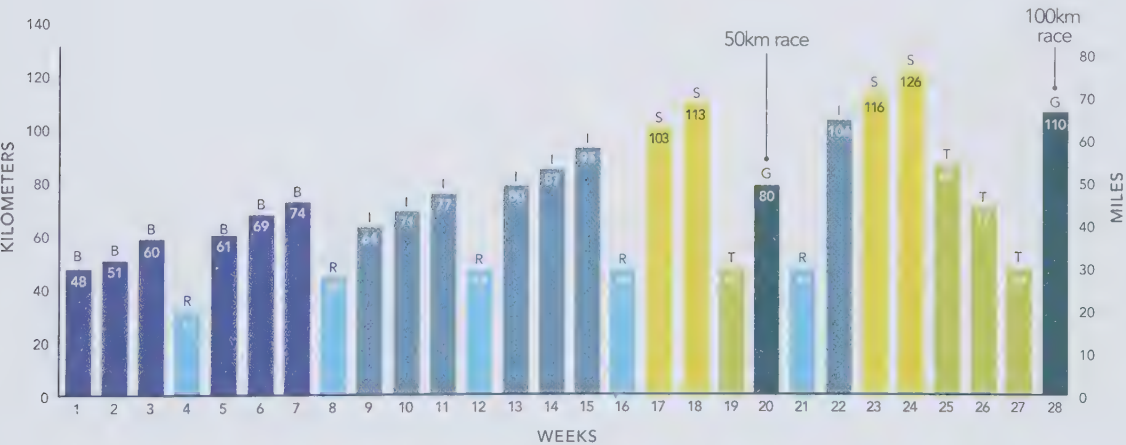


Fig. 11.8 Idealized Progression, Cat 2 Single A Priority 100K Run
This is the idealized plan for an athlete who has a single A priority running race with an intermediate B priority race. This plan could be reasonable for many Category 1 runners during the second season of structured training. It would also be a good plan for a recreational runner looking to complete a personal mountain-running project.

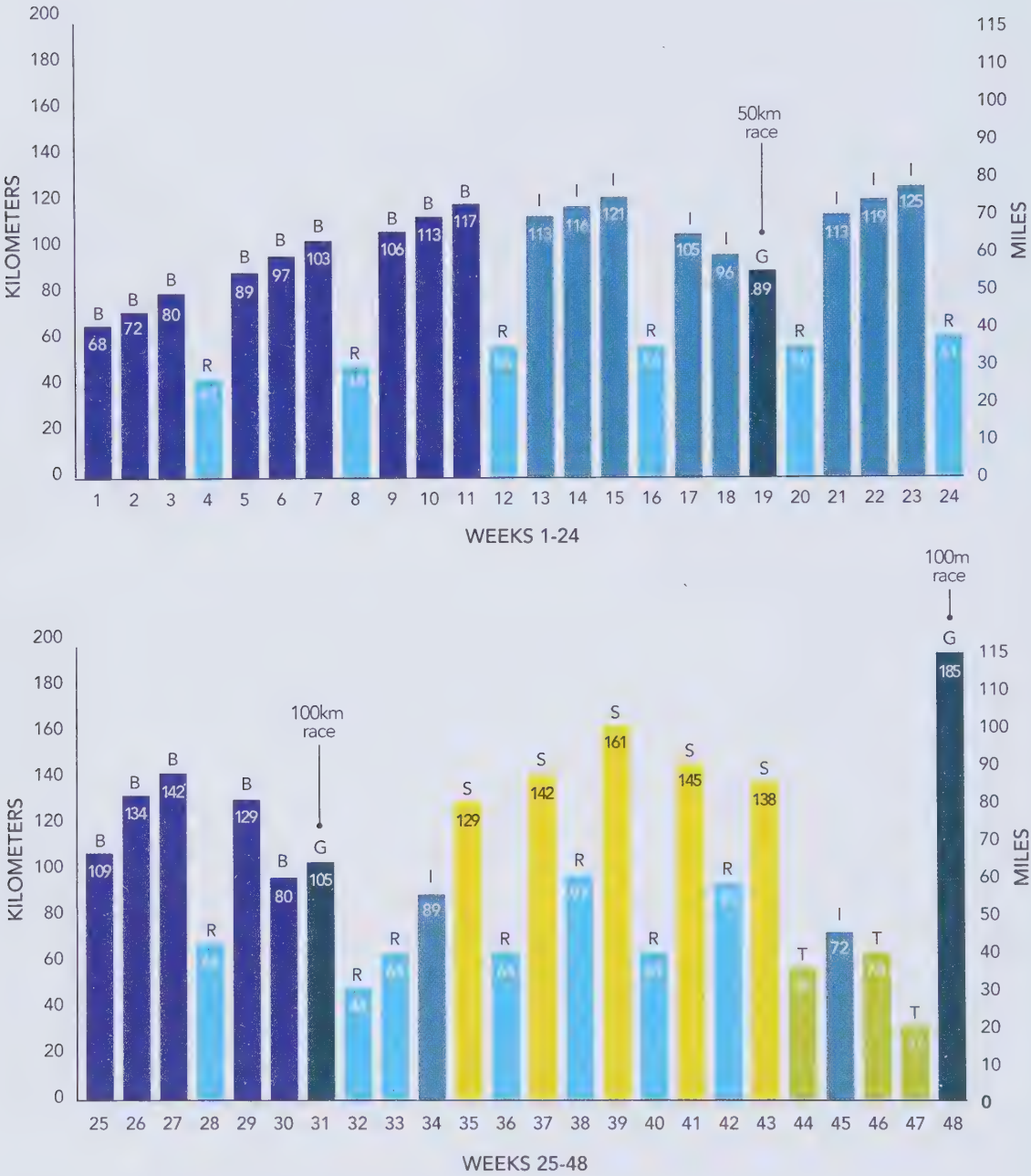


Fig. 11.9 Progression for a 100 Mile Running Race

This is an idealized progression toward a single A priority 100-mile running race with two intermediate B priority races. A plan like this is most reasonable for a Category 2 runner. Category 1 runners may be tempted to rush to these longer distances, but we strongly recommend that you build your running base over a few years of continuous training if you hope to prosper in the long term and avoid career-ending burnout and injury.

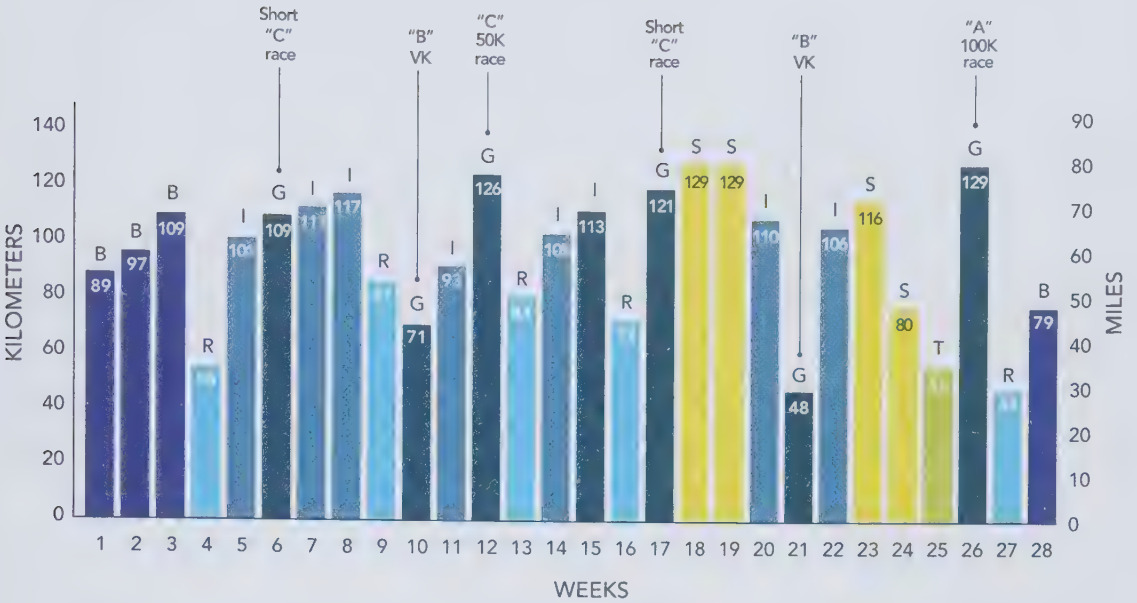


Fig. 11.10 Progression for Cat 2 Runner with a Good Base

This is how a Category 2 mountain runner might approach planning for a spring, summer, and fall of racing. Due to the athlete's high base level of fitness, none of these planned races exceed his or her current capacity. He or she will be able to use several races throughout the season as hard workouts without having to interrupt the normal flow of training. Too many athletes try to accomplish such a season with too small of a base and end up discouraged due to injury and/or exhaustion. We encourage you to take the long view and strive to reach this level over several years of steady work.

Step 3. Pick from the Week-Type Menu

Choose the week type below that matches the letter above the week in your macro-cycle plan. The workouts are listed beneath each week heading in the menu below with a description and specific recommendations for implementing the workout. Next, distribute the listed workouts for that week into your plan calendar in a way that fits your schedule using the guidelines that have been given according to the type of workout.

B (BASE) WEEK

These are the foundational weeks upon which all the other training rests. They build aerobic capacity, improving running economy and increasing your strength. The accumulation of time, distance, and vertical in the aerobic base intensity, Zones 1 and 2, is the aim. If you are aerobically deficient (experiencing ADS, see the Ten Percent Test on page 91), then you will do nearly all your aerobic

base training in Zone 2. Otherwise, you will use Zone 2 sparingly for no more than 15 percent of total weekly volume and do the base work in Zone 1.

Strength: Two workouts per week. These can be General or Specific, depending on your needs. Use the appropriate strength phase from your strength assessment. Two Core Strength workouts can be done on these or other days.

Active recovery: Include active recovery work on problem areas. Stretch, roll, massage, swim. Be proactive with self-maintenance. Do this at least three times per week.

Recovery workouts: Count this in the aerobic base distance for the week. These can be planned as very light training days after a heavy day. Or any day when you feel “off” after a warm-up and your body is telling you it is not ready to be pushed. These would be days when you would give the workout a C or D grade. Cut the intensity dramatically and shorten the workout if needed. These workouts should leave you feeling better than when you started, either immediately afterward or a few hours later. If they do not help, consider taking the next day off.

Aerobic base workouts: Start your planning by selecting one day each week for your long workout. Aim for between 30–40 percent of your overall weekly distance/vertical on this one run/ski. In lower-volume weeks, the percentage of the overall weekly volume will be higher on this one day. As your weekly volume increases, you may need to make the long day a smaller percentage of the weekly total. Distribute the remaining distance over at least four workouts. These runs will include the warm-ups and cool-downs for Hill Sprints/Bounding and ME workouts. Two of these runs should include some fast downhill running pick-ups. Skimo can also include skiing and roller skiing. Ski mountaineers optionally can include some Ski Striding. See sidebar on ski simulations (see page 313).

Muscular endurance: If muscular endurance is a weakness of yours (see page 240), this would be a good time to begin focused ME workouts. Ski mountaineers, use once per week in the early weeks.

I (INTENSITY) WEEK

Refer to the intensity guidelines for more details. These weeks introduce higher-intensity training in Zones 3 and 4, which will directly improve your endurance. They supplement the B week training. They do not replace it. If you are aerobically deficient (ADS, see the Ten Percent Test on page 91), then you will do nearly all your aerobic base training in Zone 2. Otherwise, you will drop Zone 2 and do the base work in Zone 1.



Luke Nelson, Chloë Lanthier, and Clare Gallagher run in the Dolomites, Italy. Photo: Eliza Earle

Strength: One Hill Sprint/Bounding workout these weeks. Continue two Core Strength workouts each week. Ski mountaineers, use Hill Bounding.

Active recovery: Include active recovery work on problem areas. Stretch, roll, massage, swim. Be proactive with self-maintenance. Do this at least three times per week.

Recovery workouts: Count this in the aerobic base distance for the week. Terrain should be less steep. These can be planned as very light training days after a heavy day. Or they can be unplanned and used when you feel “off” after a warm-up and your body is telling you it is not ready to be pushed. These would be days when you would give the workout a C or D grade. Cut the intensity dramatically and shorten the workout if needed. These workouts should leave you feeling better than when you started—if not immediately afterward, then a few hours later. If they do not help, consider taking the next day off.

Aerobic base workouts: Start your planning by selecting one day each week for your long workout. Aim for between 30–40 percent of your overall weekly distance/vertical on this one run/ski. In lower-volume weeks, the percentage of the overall weekly volume will be higher on this one day. As your weekly volume







Riccardo and Giulia Dallio do a snowy aerobic base workout in Val Venegia, Trentino, Italy.
Photo: Federico Modica

increases, you may need to make the long day a smaller percentage of the weekly total. Distribute the remaining distance over at least four workouts. These runs will include the warm-ups and cool-downs for Hill Sprints/Bounding and ME workouts. Two of these runs should include some fast downhill running pick-ups. For skimo, also include ski simulations. Ski mountaineers optionally can include skiing or ski simulations. See sidebar on ski simulations (see page 313).

Muscular endurance: Drop muscular endurance and begin integrating Zone 3 or Zone 4 workouts. Ski mountaineering, move to two times/week.

High intensity: For a full discussion, see page 305. Begin by adding Zone 3 intervals. Start with time in Zone 3 that's no more than 5 percent of your weekly total. Start with duration and overall volume that allows you to recover in forty-eight hours. If you still feel tired legs seventy-two hours later, then cut back on the next workout. When you can handle 10 percent of the total weekly time in Zone 3, begin to replace some Zone 3 with Zone 4. Skimo should use a ski-specific mode. Ski mountaineers, disregard this category.

R (RECOVERY) WEEK

The recovery week is intended to act as a consolidation week to allow your body to catch up. Cut to 50 percent of the total volume. The focus here needs to be on recovery first. The purpose of the recovery week or few days is not to add to your fitness bank account, but to get you recovered as well and as quickly as possible so you are ready to start building fitness again. This week should be the most loosely structured so that you can take days off or do especially light training. You want to be feeling like attacking the next week by Sunday. If not, you didn't recover. Insert alternative recovery training like swimming or easy cycling, or any other favored recovery activity. If you are aerobically deficient (ADS, see the Ten Percent Test on page 91), then do nearly all your aerobic base training in Zone 2. Otherwise, use Zone 2 sparingly for no more than 15 percent of total weekly volume and do the base work in Zone 1.

Strength: One Core Strength workout per week. Cut Hill Sprint/Bounding volume in half, doing these exercises late in the week. Ski mountaineers, use Hill Bounding cut in half.

Active recovery: Focus active recovery work on problem areas this week. Stretch, roll, massage, swim. Be proactive with self-maintenance. Do this at least three times per week.

Recovery workouts: Do at least two workouts in this intensity zone this week. Count this in the aerobic base distance for the week. Terrain should be less steep. These can be planned as very light training days after a heavy day. Or they can be unplanned and used when you feel "off" after a warm-up and your body is telling you it is not ready to be pushed. These would be days when you would give the workout a C or D grade. Cut the intensity dramatically and shorten the workout if needed. These workouts should leave you feeling better than when you started—if not immediately afterward, then a few hours later. If they do not help, consider taking the next day off.

Aerobic base workouts: Aim for a long workout of around 40 percent of your overall weekly distance/vertical. Do this late in the week when you are more recovered. These will include the warm-up and cool-down runs done in conjunction with Hill Sprints/Bounding and high intensity. Do one other workout this week at this intensity of about 20 percent of the total; do the rest in the Recovery Zone. One of these runs should include some fast downhill running pick-ups. Skimo racers can also include ski walking, roller skiing, and Moose Hoofing.

Muscular endurance: Drop or cut by 50 percent.

High intensity: Cut all high-intensity work by half from the last week. Do one workout late in the week. Skimo racers can also include ski walking, roller skiing, and Moose Hoofing. Ski mountaineers, disregard this category.

S (SPECIFICITY) WEEK

This is a week used in the later weeks of your plan to prepare you for the specific demands of your A priority event. For ultrarunners and ski mountaineers this will come from doing back-to-back long runs/skis. This gives you a similar training effect as doing one very long day by mimicking some of the demands of your event, but it reduces the stress and shortens the recovery time significantly.

For shorter, high-intensity events like a VK or skimo race, this comes from reducing the aerobic base training and adding in more Zone 4 interval work. If you are aerobically deficient (ADS, see the Ten Percent Test on page 91), then do nearly all your aerobic base training in Zone 2. Otherwise, use Zone 2 sparingly for no more than 15 percent of your total weekly volume and do the base work in Zone 1.

Strength: One Hill Sprint/Bounding workout these weeks. Continue two Core Strength workouts each week.

Muscular endurance: Ski mountaineers, continue the ME workout once per week. For runners and skimo racers cut so that the high intensity can be higher quality.

Active recovery: Include active recovery work on problem areas. Stretch, roll, massage, swim. Be proactive with self-maintenance. Do this at least three times per week.

Recovery workouts: Do at least two workouts in this intensity zone this week. Count this in the aerobic base distance for the week. Terrain should be less steep. These can be planned as very light training days after a heavy day. Or they can be unplanned when you feel “off” after a warm-up and your body is telling you it is not ready to be pushed. These would be days when you would give the workout a C or D grade. Cut the intensity dramatically and shorten the workout if needed. These workouts should leave you feeling better than when you started—if not immediately afterward, then a few hours later. If they do not help, consider taking the next day off.

Aerobic base workouts: If training for a long event, do back-to-back long workouts these weeks. Aim for between 60–80 percent of your overall weekly distance/vertical on these two workouts. Distribute the remaining distance over at least three workouts. These will include the warm-ups and cool-downs for Hill Sprints/Bounding and ME workouts. Two of these workouts should include some fast downhill running or skiing pick-ups. Skimo racers can also include Ski Striding, roller skiing, and Moose Hoofing.

If training for normal skimo shorter runs like VK, double up on the high-intensity workouts.

High intensity: For ultradistance runners, cut all Zone 3 and do only one Zone 4 interval session at no more than the same volume as the previous week. Do this workout separated by at least 72 hours from the back-to-back long runs.

For VK or skimo racers, cut all Zone 3 and add a second Zone 4 interval session separated by at least 72 hours from the previous one separated by recovery workouts. The total time in Zone 4 for the week should be no more than 15 percent of the previous week.

T (TAPER) WEEK

Tapering is a very individual preference with some top athletes like Kilian preferring not to taper for races. This is especially true if an athlete is racing often. In that case, too much tapering would result in a drop in fitness over the season. Traditionally the load would be reduced for some days or even weeks (in the case of very long races) before the event so that the athlete is fresh on the day that counts. How long and how sharply you drop the training load will best be learned through experience. Our example plans will include traditional taper periods. In general, A priority races will deserve a more dramatic taper than B priority races. C priority events will be treated as normal workouts. The training is over once you enter a taper period. You should not try to sneak in some special or new workout in hopes of getting fitter or faster. The time for that is long past. You want to gradually reduce the training load as little as necessary to allow your legs to refresh without losing fitness. That's a tricky balancing act.

Strength: One Core Strength workout per week. Cut Hill Sprint/Bounding by 30 percent from week to week.

Active recovery: Focus on active recovery, work on problem areas this week. Stretch, roll, massage, swim. Be proactive with self-maintenance. Do this at least three times per week.

Recovery workouts: You will be cutting overall training volume during the taper and all but two recovery sessions will be in this intensity range.

Aerobic base workouts: Aim for a long run or ski of around 50 percent of your overall weekly distance/vertical. These will include the warm-up and cool-down done in conjunction with Hill Sprints and high intensity. One other Zone 2 workout with no more than 10 percent of total weekly distance/vertical. Include a few short fast downhill runs. Skimo racers, include a few quick-tempo pick-ups.

High intensity: Keep all high-intensity work short with long rests. Overall volume of intensity should drop about 30 percent from week to week. This is just for sharpening.

G (GOAL) WEEK

This is the race week itself. It is part of the taper period but also a time to take good care of yourself. This week may involve travel. Nothing you do this week will make you faster, but many things you could do will make you slower. Rest and relax but be active each day. Too much rest will make you sluggish.

Strength: No strength this week.

Active recovery: Focus on active recovery; work on problem areas this week. Stretch, roll, massage, swim. Be proactive with self-maintenance. Do this at least three times per week.

Recovery workouts: Do only Recovery Zone this week. No more than four days, just to keep the legs loose.

High intensity: Spice two of your workouts with three or four accelerations up to a fun, fast pace for thirty seconds each with three minutes of easy jogging between. None for ski mountaineers.

Base Period Guidelines

Keep the following principles in mind as you do the heavy lifting of planning. Progression of the training load is one of the most important elements to consider as you embark on this base-building phase of what may be your first structured training plan. Increase the load too fast and, at best, you'll get tired and need an unplanned break to recover. At worst, you'll get injured or become overtrained, which could derail your plan for weeks and even months. Getting



Jacqui Levy and Alyson Dimmitt Gnam, determined to keep their training volume on track, transition to spikes during a ski tour outside of Missoula, Montana. Photo: Steven Gnam

this right means selecting an appropriate weekly vertical and progressing at a controlled rate. No one can tell you what is an appropriate amount of starting vertical. We have suggested some guidelines, but this will depend on all the factors that give rise to the individual response to training: age, training history, genetics, and life stressors. While the weekly vertical progressions shown in the charts are conservative and should prevent overuse injuries and overtraining, do not proceed stubbornly if your body is telling you something is wrong. Listen to your body.

Note: If you are training at well below 400 hours/year, then expect to spend additional training cycles in this base-building phase of training. This applies especially to juniors and the inexperienced. Do not skip to the Category 2 Athlete section too early or you will struggle to handle the workload and may not see improvement.

Progression of the Training Volume (Weekly Vertical and Distance)

While we made some general weekly distance recommendations a few pages ago, below are a few ideas to keep in mind as you plan.

Guidelines for progressing the training load:

- For the smoothest long-term progression, keep distance, vertical, and time increases at 10 percent or less from week to week except after a recovery week, when you may increase these much more.
- Do not increase distance and vertical for more than three weeks in a row.
- Do not increase distance, vertical, *and* intensity in a single week. Try to increase only one at a time. If you need to increase two of these, then keep the increases to under 15 percent. Example: 10 percent distance increase with 5 percent vertical increase, or 8 percent increase in Zone 3 time with 7 percent increase in distance.
- Do not make distance/vertical increases of greater than 15 percent for two weeks in a row.
- Typically you should follow large jumps (near 15 percent) in weekly distance, vertical, or intensity with a recovery week to allow your body to adapt.

Modulation of the Training Load

In our macroplan examples, we mostly used three weeks of gradually building the workload followed by a recovery week, during which the training load is greatly reduced. The use of a 3:1 build: recovery cycle is not meant to impose an inflexible structure. You may find that your body does best with a 2:1 cycle. This will take some experimentation and you may need to vary this depending on how you are adapting to the training. Do not stubbornly follow any plan without paying attention to how your body is adapting to it (see the overtraining discussion on page 136). Most importantly: You need to build modulation into your plan. Rest *before* you are forced to. Recovery after hard work is when the gains in fitness occur. During recovery weeks, take enough recovery days so that you feel fresh by the end.

Guidelines for modulating the training load:

- Pay attention to your body. Keep a log. Experiment to find what works for you.
- Modulate the vertical/distance in a manner similar to the way the macroplan weeks show. As much as possible, stick to a progression and don't jump around randomly.

- Recovery weeks should drop the training load typically at least 40 to 60 percent of the last big week's volume. Don't do more than one intensity session.
- If you do not have much experience, especially *successful* experience, with high-intensity sessions, you may need to shift to a 2:1 or even 1:1 weekly cycle to allow adequate recovery.
- If you intentionally use an overreaching period (see page 99), then plan for a correspondingly lighter recovery period, dropping the load by as much as 70 percent.

Vertical

Just as with distance, the amount of vertical feet (meters) you include in your weekly training is going to depend on several factors: the vertical in your goal event, your access to hills and mountains for training, and your training history. For best results, pick your goals according to what you have available to train on, and then train on hills similar to the goal race. Just as with weekly distance, if you are new to these sports, you'll need to start small and build vertical gradually over the whole plan. Whereas if you have a good history of training, you'll start with and maintain a bigger weekly vertical.

Guidelines for weekly vertical:

- If you are new to these sports, start with no more than a third to one-half of your goal event's total vertical in your first week.
- Plan on training a minimum of your goal event's total vertical during two of your late Base Period weeks.
- Main workouts should be done on foot: running, ski simulations, roller skiing, or skiing.
- These big vertical weeks should correspond to the longer distance weeks.
- Cut vertical by a similar amount as distance reduction during recovery weeks (40–60 percent).

High Intensity

While it is fairly easy to prescribe low-intensity training volumes (because more is almost always better), it is much more challenging to prescribe specific high-intensity workouts. The principle of individuality comes into play much more so when speaking about intensity. Your training history will affect how often and how much high-intensity work is best for you to incorporate into each session. If you have a history of high-intensity training, you will understand what an appropriate intensity dosage is as well as being able to recover better from these workouts. If you are new to adding high-intensity workouts into a high-volume base-building

period, you will need to be more conservative as you learn how much and how often you can handle them while maintaining a high aerobic volume.

Skimo: High-intensity workouts play an important role in skimo training because the races are generally performed at significantly higher intensities than ultradistance running events. Regardless, high intensity is still a supplement to, and not a replacement for, the basic aerobic work you need. Especially for Category 1 athletes approaching their first skimo season, aerobic (Zones 1–2) vertical base workouts should take priority over intensity. Ideally you will not add high-intensity workouts until your AeT heart rate or pace is within 10 percent of your LT heart rate, but that is not always practical. (See the Ten Percent Test on page 91.)

Runners: If you are competing at distances for which you have adequate base volume, then adding high intensity will be a valuable tool for increasing performance. However, if you are moving up to new distance and stepping into new territory, you'd be wise to prioritize increasing the distance and vertical before adding intensity.

The practical reality is that at some point you must begin to add intensity regardless of the state of your aerobic capacity. Begin to add one Zone 3 workout each week when within two months of your first race. Adding intensity should not require reducing the volume of training more than 5 percent. If it does, you risk losing the aerobic base you have built. Balancing the introduction of intensity into your program is, like most aspects of training, highly individual and can be tricky to do well. Following are some guidelines to help you.

Guidelines for intensity:

- Kílian's preferred way of introducing high intensity is with Zone 3. Include one Zone 3 workout per week when you feel like you are handling the Zones 1–2 volume well. While a highly trained athlete may be comfortable with two sessions per week totaling as much as one to two hours in Zone 3, it will take years to build up to this volume. The best practice for Category 1 athletes will be to ease into Zone 3 with one workout a week with a total of twenty to thirty minutes of Zone 3 done as an interval workout such as 3x7 or 3x8 min with short, active rest intervals in a 3:1 or 4:1 work-to-rest ratio. If you do not recover well from this workout, then either you were pushing too hard or the overall volume of work was too high. Adjust accordingly and try again.
- The volume of high-intensity training in each week, and within each workout, should increase gradually and not start out too high. Add only as much as you can before it forces you to drop the volume of your aerobic base (Zones 1–2) training by more than 5 percent. If you can't maintain your volume of Zones 1–2 training, you risk having a short peak followed by a drop in performance.

- The type of intensity you add to your program is dependent upon the event you are targeting. Long Zone 3 sessions will serve well for long-distance races. On the other hand, shorter Zone 4 and Zone 5 sessions will be very useful when you are training for a VK or shorter race.
- Ease into intensity training by first adding one weekly Zone 3 workout. Start with about 5 percent of weekly volume in Zone 3. Add Zone 3 volume gradually and not more than 10 percent per week.
- Kilian recommends increasing high intensity volume only when you are comfortable handling the current volume/intensity mix. Adding intensity too fast is one of the most common mistakes athletes make.
- Drop or reduce Zone 2 training when you begin adding Zone 3 or 4 work to your plan. Shift to more aerobic base work in Zone 1.
- Consider the inclusion of Zone 4 intervals to your week only when the volume of Zone 3 has reached about 10 percent of your weekly volume and you are feeling comfortable with that amount—recovering well and not seeing any drop in overall weekly volume.
- Add Zone 4 intervals by gradually replacing the volume of Zone 3 with Zone 4 work done each week. Start with no more than 5 percent of your total weekly volume in Zone 4. If you must drop the overall weekly Zones 1–2 volume by more than 5 percent, just use the Zone 3 more until you feel that is easy to handle.
- If you are new to structured training, start your Zone 4 intervals by doing 30/30s (see page 177) before progressing to longer and more taxing intervals.

Strength

Keep in mind that strength for the sake of strength is not the point. Strength in endurance sports is used only in a way that improves performance and prevents injury. It plays a supportive role. The most effective form of strength training for these sports is muscular endurance. There are a number of ways to improve this quality, which translates directly into improved performance.



ATHLETE STORIES

The Ghost on the Mountain

Luke Nelson

Rumors on the mountain were spreading. Word among the groomers was that there had to be something on the mountain. Even the long-time cat drivers had expressed concern. I overheard the conversation while eating breakfast in the lodge after a long training session.

"New footsteps appeared each time I passed the steep section under the Sunshine Chair," one groomer said. "Four laps, four different sets of footprints."

Another groomer added, "Footsteps that start out of the blue and then just disappear—so weird. I know it has to be a ghost or something."

I chuckled as I scarfed down hash browns and eggs. It wasn't the first time this season that I had caught wind that the night grooming staff was concerned about a ghost on the mountain. You would think that since I trained late at night or early in the morning, often in the dark, I too would be concerned about the ghost.

But I knew better. The ghost was me.

I had committed to putting in a very focused training block on snow in order to prepare for the upcoming US Ski Mountaineering Championships, and because I had recently started a new job that occupied most daylight hours, I trained in the dark. I trained every day, and most days twice a day. I would park as low on the mountain as possible in order to maximize vert, which put my truck out of sight of the grooming staff. The majority of my training occurred well off of the groomed runs and kept the beam of my headlamp hidden.

Once or twice a week I would do intervals that included a transition from skinning to booting up a steep, groomed run. I would time it so that I would literally race the groomers up the mountain. The competition helped me push the interval harder than I would have been able to otherwise. Because I started the interval skinning and then transitioned to booting, my tracks weren't obvious until I was kicking steps. Well before the cat turned around, I would transition back to skiing and slip into the woods to ski back to the bottom.

Once I heard about the ghost, I may have gone to extra measures to assure that I didn't get spotted during my nightly training ... just to keep the story interesting.

This went on for well over a month until a week or so before the US Championships, when I had planned a time trial up the entire ski area. I popped out of the woods next to the groomer, who was very surprised to see me, and then proceeded to race him to the top of the mountain. It was a close race, but in the end the machine pulled ahead. The cat driver jumped out at the top and waited for me to crest the final climb. He let out a whoop and gave me a high-five, then told me I owed him a beer for how nervous he had gotten about the ghost on the mountain—the ghost he was relieved to see was just me.

A week later I stood on top of the podium in Jackson Hole. The consistent training, even in the dark of night, had paid off. ■

Luke Nelson running the ninety-three mile-long Greys River Valley. It was minus twenty Fahrenheit when the team started and it stayed cold for the entire trip. Wyoming. Photo: Fredrik Marmsater

